PROJECT 3: SQL DATA ANALYSIS

E-Commerce Database Query Analysis

Industrial Training Kit | Batch: 2026 | DecodeLabs

1. Problem Statement

Core Question: How can SQL queries be used to extract actionable business intelligence from the e-commerce database? This project demonstrates proficiency in writing structured queries to filter, group, and aggregate raw data into meaningful insights.

Key Objectives:

Extract specific data using WHERE filters

Calculate aggregations using GROUP BY

Filter grouped data using HAVING

Sort and present data using ORDER BY

2. Methodology

Database Details:

Platform: SQL Server Management Studio (SSMS)

Table Name: sales

Columns: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, PaymentMethod, OrderStatus, ReferralSource, CouponCode, TotalPrice

Total Records: 1,200 orders

Analytical Approach: WHERE filters to isolate records, GROUP BY with SUM/COUNT/AVG for summaries, HAVING to filter aggregated groups, ORDER BY for presentation, and PERCENTILE_CONT for the five-number summary.

3. Key Queries and Findings

3.1 Basic Filtering

Filter	Result
TotalPrice > $500	634 orders (52.83% of total)
OrderStatus = 'Cancelled'	250 orders (20.83% of total)
TotalPrice > $1,000 AND PaymentMethod = 'Credit Card'	87 orders

Note: The Cancelled count is corrected to 250 to match the verified order status breakdown in Section 3.3 below (the originally reported 229 did not match the underlying pivot table data).

3.2 Revenue by Product

```
SELECT product, COUNT(*) AS total_orders, SUM(totalprice) AS revenue FROM sales GROUP BY product ORDER BY revenue DESC;
```

Product	Revenue
Chair	$195,620.11
Printer	$195,612.61
Laptop	$192,126.56
Tablet	$186,568.95
Monitor	$175,651.41
Desk	$167,459.93
Phone	$151,722.39
TOTAL	$1,264,761.96

Note: Printer was missing from earlier drafts of this table; it is a real product in the sales table and is included here, bringing the total from 6 to 7 products. Order-level counts (total_orders) are not yet confirmed against verified output and have been omitted pending confirmation; only revenue, which is confirmed, is shown.

Key Finding: Chair generates the highest revenue ($195,620.11), with Printer a close second ($195,612.61). Phone generates the least.

3.3 Order Status: Count and Average Value

```
SELECT orderstatus, COUNT(*) AS order_count, AVG(totalprice) AS avg_order_value FROM sales GROUP BY orderstatus ORDER BY avg_order_value DESC;
```

Order Status	Order Count	Avg Order Value
Cancelled	250	$1,105.58
Pending	237	$1,081.55
Delivered	231	$1,050.22
Shipped	235	$1,047.49
Returned	247	$984.93

TOTAL	1,200	—

Key Finding: Cancelled orders actually carry the highest average order value ($1,105.58), not the lowest as previously reported — customers cancelling tend to be cancelling larger orders, which is worth investigating directly (e.g., checkout friction or payment failures on higher-value carts). Returned orders have the lowest average value ($984.93).

3.4 Revenue Excluding Cancelled Orders

```
SELECT product, SUM(totalprice) AS total_revenue, COUNT(OrderID) AS order_count FROM
sales WHERE orderstatus != 'cancelled' GROUP BY product HAVING SUM(totalprice) > 5000
ORDER BY total_revenue DESC;
```

Product	Revenue (Excl. Cancelled)	Order Count
Printer	$156,937.45	146
Laptop	$148,364.84	138
Chair	$146,959.13	133
Tablet	$145,938.99	145
Monitor	$138,593.97	128
Desk	$127,872.24	135
Phone	$123,699.13	125

Key Finding: Printer leads revenue once cancelled orders are excluded, overtaking Chair, which led on total (including cancelled) revenue. This suggests Chair carries a disproportionate share of cancellations relative to Printer.

3.5 Monthly Revenue Trend

```
SELECT DATENAME(month, date) AS months, SUM(totalprice) AS total_revenue FROM sales
GROUP BY DATENAME(month, date), MONTH(date) ORDER BY MONTH(date);
```

Month	Revenue
January	$124,313.23
February	$112,344.78
March	$123,840.93
April	$109,186.05
May	$135,142.59

June	$170,616.13
July	$85,784.64
August	$86,343.21
September	$69,321.65
October	$89,834.82
November	$75,493.43
December	$82,540.50

Key Finding: June is the peak revenue month ($170,616.13), not January. Revenue is consistently higher across January–June than across July–December, with September the weakest month ($69,321.65).

3.6 Five-Number Summary

```
WITH Stats AS (   SELECT totalprice,     PERCENTILE_CONT(0.25) WITHIN GROUP (ORDER BY totalprice) OVER() AS Q1,     PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY totalprice) OVER() AS Median,     PERCENTILE_CONT(0.75) WITHIN GROUP (ORDER BY totalprice) OVER() AS Q3   FROM sales ) SELECT MIN(totalprice) AS Minimum, MAX(Q1) AS Q1, MAX(Median) AS Median,        MAX(Q3) AS Q3, MAX(totalprice) AS Maximum FROM Stats;
```

Metric	Value
Minimum	$11.39
Q1 (25%)	$410.52
Median (50%)	$823.62
Q3 (75%)	$1,578.48
Maximum	$3,456.40

Interpretation: The distribution is right-skewed (Mean of $1,053.97 exceeds the Median of $823.62), confirming that a minority of high-value orders pull the average upward.

4. Key Performance Indicators (KPIs)

```
SELECT COUNT(*) AS Total_Orders, SUM(totalprice) AS Total_Revenue, AVG(totalprice) AS Average_Order_Value,        (COUNT(CASE WHEN orderstatus IN ('Delivered','Shipped') THEN 1 END) * 100.0 / COUNT(*)) AS Fulfillment_Rate FROM sales;
```

#	KPI	Value
1	Total Orders	1,200

2	Total Revenue	$1,264,761.96
3	Average Order Value (AOV)	$1,053.97
4	Order Fulfillment Rate	≈39% (38.83%)

5. The Alias Trap (Execution Order Note)

SQL evaluates WHERE before SELECT, so a column alias defined in SELECT cannot be referenced in WHERE within the same query:

```
-- WRONG: alias "rev" does not exist yet at WHERE-evaluation time SELECT TotalPrice * 1.1 AS rev FROM sales WHERE rev > 1000;  -- CORRECT: repeat the full expression SELECT TotalPrice * 1.1 AS rev FROM sales WHERE TotalPrice * 1.1 > 1000;
```

6. Recommendations

Investigate Cancellations on High-Value Orders: Cancelled orders have the highest average value ($1,105.58) of any status, meaning the business is losing its larger transactions disproportionately — review checkout flow and payment failure points for high-cart-value sessions specifically.

Address the Chair Cancellation Gap: Chair leads on total revenue but drops to third once cancelled orders are excluded, while Printer moves into first — this points to a product-specific issue (stock, pricing, or fulfillment) worth a closer look.

Investigate the Second-Half Revenue Drop: revenue is consistently lower from July through December than from January through June, with September the weakest month ($69,321.65) — determine whether this is seasonal, a marketing spend change, or a data gap.

Monitor Outliers: the lowest recorded order ($11.39) and highest ($3,456.40) sit far apart from the Q1–Q3 range of $410.52–$1,578.48; flag extreme low-value orders for review as potential pricing errors.

7. Conclusion

SQL proved effective for extracting business intelligence from the e-commerce database through filtering, grouping, aggregating, and sorting. The strongest signal in this analysis is that Cancelled orders carry the highest average value of any status, meaning cancellations are concentrated among the business's most valuable transactions rather than spread evenly. Revenue also splits cleanly into a stronger January–June period and a weaker July–December period, peaking in June rather than January. Together with the fulfillment rate of 38.83%, these findings point to a business that is generating strong top-line revenue but losing a meaningful share of its highest-value orders before they complete.

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